Supplementary Material

8 Convergence analysis

In this section, we prove all the theorems in Section 5.

8.1 proof of Theorem 1

Proof We first define $vec(\mathcal{X})$ as the vectorization of $\mathcal{X}$ gives a vector, which is obtained by stacking all mode-1 fibers of $\mathcal{X}$.

Next, from Eq. (6), we know that

$$vec(\nabla_{\mathcal{G}} H(\bar{\mathcal{G}}, A_{(N)}) - \nabla_{\mathcal{G}} H(\mathcal{G}, A_{(N)})) = vec((\bar{\mathcal{G}} - \mathcal{G})(\otimes_{i=N}^{1}((A_i)^T A_i))).$$

Since $\|\mathcal{G}\|_F = \|\text{vec}(\mathcal{G})\|_F$, thus we infer

$$\begin{aligned}\|\nabla_{\mathcal{G}} H(\bar{\mathcal{G}}, A_{(N)}) - \nabla_{\mathcal{G}} H(\mathcal{G}, A_{(N)})\|_F &= \|vec((\bar{\mathcal{G}} - \mathcal{G})(\otimes_{i=N}^{1}((A_i)^T A_i)))\|_F \\ &\leq \| \otimes_{i=N}^{1} ((A_i)^T A_i)\|_F \|vec(\bar{\mathcal{G}} - \mathcal{G})\|_F \\ &= L_{\nabla_{\mathcal{G}} H} \|\bar{\mathcal{G}} - \mathcal{G}\|_F,\end{aligned}$$

and from Eq. (6), we also have

$$\begin{aligned}&\|\nabla_{A_n} H(\mathcal{G}, \{A_j\}_{j=1}^{n-1}, \bar{A}, \{A_j\}_{j=n+1}^{N}) - \nabla_{A_n} H(\mathcal{G}, A_{(N)})\|_F \\ &\leq L_{\nabla_{A_n} H} \|\bar{A} - A_n\|_F,\end{aligned}$$

where $B_n = G^{(n)}(\otimes_{i=N, i\neq n}^{1} A_i)^T$ and

$$L_{\nabla_{A_n} H} = \begin{cases} \|B_n (B_n)^T\|_F + \| \sum_{q=1}^{Q} \alpha_q L_q\|_F, & \text{if } n = N, \\ \|B_n (B_n)^T\|_F, & \text{else.} \end{cases}$$

Thus, Theorem 1 is true. □

8.2 proof of Theorem 2

Proof (i) If step 9 of Algorithm 1 is not true, then the Theorem 2(i) is true, if step 9 of Algorithm 1 is true, since Theorem 1 and Proposition 4, we know that

$$\begin{aligned}H(\mathcal{G}^{k+1}, A_{(N)}^k) \leq \langle \nabla_{\mathcal{G}} H(\mathcal{G}^k, A_{(N)}^k), \mathcal{G}^{k+1} - \mathcal{G}^k \rangle + H(\mathcal{G}^k, A_{(N)}^k) \\ + \frac{L_{\nabla_{\mathcal{G}^k} H}}{2} \|\mathcal{G}^{k+1} - \mathcal{G}^k\|_F^2.\end{aligned} \tag{9}$$

From Eq. (7), we obtain

$$F_G(\mathcal{G}^k) \geq F_G(\mathcal{G}^{k+1}) + \frac{1}{2\sigma_G^k} \|\mathcal{G}^{k+1} - \mathcal{G}^k\|_F^2 + \langle \nabla_{\mathcal{G}} H(\mathcal{G}^k, A_{(N)}^k), \mathcal{G}^{k+1} - \mathcal{G}^k \rangle. \tag{10}$$

Then sum of the Eq. (9) and Eq. (10), we have

$$\begin{aligned}H(\mathcal{G}^{k+1}, A_{(N)}^k) \leq H(\mathcal{G}^k, A_{(N)}^k) + F_G(\mathcal{G}^k) - F_G(\mathcal{G}^{k+1}) \\ - (\frac{1}{2\sigma_G^k} - \frac{L_{\nabla_{\mathcal{G}^k} H}}{2}) \|\mathcal{G}^{k+1} - \mathcal{G}^k\|_F^2.\end{aligned} \tag{11}$$

Similarly, since Theorem 1 and Proposition 4, we know that

$$H(\mathcal{G}^{k+1}, A_1^{k+1}, \{A_i^k\}_{i=2}^N) \leq \langle \nabla_{A_1} H(\mathcal{G}^{k+1}, A_{(N)}^k), A_1^{k+1} - A_1^k \rangle + H(\mathcal{G}^{k+1}, A_{(N)}^k) + \frac{L_{\nabla_{A_1^k} H}}{2} \|A_1^{k+1} - A_1^k\|_F^2. \tag{12}$$

From Eq. (7), we can also obtain

$$F_1(A_1^k) \geq F_1(A_1^{k+1}) + \frac{1}{2\sigma_1^k} \|A_1^{k+1} - A_1^k\|_F^2 + \langle \nabla_{A_1} H(\mathcal{G}^{k+1}, A_{(N)}^k), A_1^{k+1} - A_1^k \rangle. \tag{13}$$

Then sum of the Eq. (12), Eq. (13) and Eq. (11), we have

$$\begin{aligned} H(\mathcal{G}^{k+1}, A_1^{k+1}, \{A_i^k\}_{i=2}^N) \leq\ & H(\mathcal{G}^k, A_{(N)}^k) + F_1(A_1^k) + F_G(\mathcal{G}^k) - F_1(A_1^{k+1}) - F_G(\mathcal{G}^{k+1}) \\ & - \rho \|A_1^{k+1} - A_1^k\|_F^2 \\ & - \rho \|\mathcal{G}^{k+1} - \mathcal{G}^k\|_F^2, \end{aligned} \tag{14}$$

where $\rho = \min(\frac{1}{2\sigma_G^k} - \frac{L_{\nabla_{\mathcal{G}^k} H}}{2}, \frac{1}{2\sigma_1^k} - \frac{L_{\nabla_{A_1^k} H}}{2})$. Assuming Theorem 2(i) holds when $n = N-1$, i.e.,

$$\begin{aligned} H(\mathcal{G}^{k+1}, A_{(N-1)}^{k+1}, A_N^k) \leq\ & H(\mathcal{G}^k, A_{(N)}^k) + F_G(\mathcal{G}^k) + \sum_{i=1}^{N-1} F_i(A_i^k) - \sum_{i=1}^{N-1} F_i(A_i^{k+1}) - F_G(\mathcal{G}^{k+1}) \\ & - \rho \|A_{(N-1)}^{k+1} - A_{(N-1)}^k\|_F^2 \\ & - \rho \|\mathcal{G}^{k+1} - \mathcal{G}^k\|_F^2, \end{aligned} \tag{15}$$

where $\rho = \min(\frac{1}{2\sigma_G^k} - \frac{L_{\nabla_{\mathcal{G}^k} H}}{2}, \{\frac{1}{2\sigma_i^k} - \frac{L_{\nabla_{A_i^k} H}}{2}\}_{i=1}^{N-1})$. Similarly, from Eq. (7), we infer

$$F_N(A_N^k) \geq F_N(A_N^{k+1}) + \frac{1}{2\sigma_N^k} \|A_N^{k+1} - A_N^k\|_F^2 + \langle \nabla_{A_N} H(\mathcal{G}^{k+1}, A_{(N-1)}^{k+1}, A_N^k), A_N^{k+1} - A_N^k \rangle. \tag{16}$$

From Theorem 1 and Proposition 4, we also have

$$\begin{aligned} H(\mathcal{G}^{k+1}, A_{(N)}^{k+1}) \leq\ & H(\mathcal{G}^{k+1}, A_{(N-1)}^{k+1}, A_N^k) \\ & + \langle \nabla_{A_N} H(\mathcal{G}^{k+1}, A_{(N-1)}^{k+1}, A_N^k), A_N^{k+1} - A_N^k \rangle \\ & + \frac{L_{\nabla_{A_N^k} H}}{2} \|A_N^{k+1} - A_N^k\|_F^2. \end{aligned} \tag{17}$$

Then sum of the Eq. (15), Eq. (16) and Eq. (17), we have

$$J(\mathcal{G}^{k+1}, A_{(N)}^{k+1}) \leq J(\mathcal{G}^k, A_{(N)}^k) - \rho \|z^{k+1} - c^k\|_F^2, \tag{18}$$

where $\rho = \min(\frac{1}{2\sigma_G^k} - \frac{L_{\nabla_{\mathcal{G}^k} H}}{2}, \{\frac{1}{2\sigma_i^k} - \frac{L_{\nabla_{A_i^k} H}}{2}\}_{i=1}^{N})$. This shows the Theorem 2(i) is true.

(ii) We know that $\frac{\alpha_q}{2} \operatorname{Tr}((A_N)^T L_q A_N)$ in Eq. (4) is nonnegative. Hence, Eq. (4) is nonnegative. From Eq. (18), we have

$$\rho \|z^{k+1} - c^k\|_F^2 \leq J(\mathcal{G}^k, A_{(N)}^k) - J(\mathcal{G}^{k+1}, A_{(N)}^{k+1}).$$

Sum of both side, since Eq. (4) is nonnegative, thus we have

$$\begin{aligned}\rho\sum_{k=1}^{\infty}\|z^{k+1}-c^k\|_F^2 &\leq \sum_{k=1}^{\infty}(J(\mathcal{G}^k,A_{(N)}^k)-J(\mathcal{G}^{k+1},A_{(N)}^{k+1}))\\ &= J(\mathcal{G}^1,A_{(N)}^1)-\inf J\\ &< \infty.\end{aligned}$$

It follows that

$$\lim_{k\to\infty}\|z^{k+1}-c^k\|_F^2=0.$$

□

8.3 proof of Lemma 1

Proof By Proposition 2 and Proposition 3, Eq. (7) follows that

$$0\in\partial_{\mathcal{G}}F_G(\mathcal{G}^{k+1})-\frac{1}{\sigma_G^k}(\mathcal{Y}^k-\mathcal{G}^{k+1})+\nabla_{\mathcal{G}}H(\mathcal{Y}^k,A_{(N)}^k). \tag{19}$$

Thus we infer

$$\begin{aligned}f_G^{k+1} &= \nabla_{\mathcal{G}}H(\mathcal{G}^{k+1},A_{(N)}^k)-\nabla_{\mathcal{G}}H(\mathcal{Y}^k,A_{(N)}^k)+\frac{1}{\sigma_G^k}(\mathcal{Y}^k-\mathcal{G}^{k+1})\\ &\in \nabla_{\mathcal{G}}H(\mathcal{G}^{k+1},A_{(N)}^k)+\partial_{\mathcal{G}}(F_G(\mathcal{G}^{k+1})+\sum_{i=1}^{N}F_i(A_i^k))\\ &= \partial_{\mathcal{G}}J(\mathcal{G}^{k+1},A_{(N)}^k). \end{aligned}\tag{20}$$

Similarly, by Proposition 2 and Proposition 3, Eq. (7) follows that

$$0\in\partial_{A_i}F_i(A_i^{k+1})-\frac{1}{\sigma_i^k}(Y_i^k-A_i^{k+1})+\nabla_{A_i}H(\mathcal{G}^{k+1},\{A_j^{k+1}\}_{j=1}^{i-1},Y_i^k,\{A_j^k\}_{j=i+1}^{N}). \tag{21}$$

Thus we also have

$$\begin{aligned}f_i^{k+1} &= \nabla_{A_i}H(\mathcal{G}^{k+1},\{A_j^{k+1}\}_{j=1}^{i},\{A_j^k\}_{j=i+1}^{N})-\frac{1}{\sigma_i^k}(A_i^{k+1}-Y_i^k)\\ &-\nabla_{A_i}H(\mathcal{G}^{k+1},\{A_j^{k+1}\}_{j=1}^{i-1},Y_i^k,\{A_j^k\}_{j=i+1}^{N})\\ &\in \nabla_{A_i}H(\mathcal{G}^{k+1},\{A_j^{k+1}\}_{j=1}^{i-1},A_i^{k+1},\{A_j^k\}_{j=i+1}^{N})\\ &+\partial_{A_i}(F_G(\mathcal{G}^{k+1})+\sum_{j=1}^{i}F_j(A_j^{k+1})+\sum_{j=i+1}^{N}F_j(A_j^k))\\ &= \partial_{A_i}J(\mathcal{G}^{k+1},\{A_j^{k+1}\}_{j=1}^{i-1},A_i^{k+1},\{A_j^k\}_{j=i+1}^{N}). \end{aligned}\tag{22}$$

Since Theorem 1, we have

$$\begin{aligned}\|f_G^{k+1}\|_F &= \|\nabla_{\mathcal{G}}H(\mathcal{G}^{k+1},A_{(N)}^k)+\frac{1}{\sigma_G^k}(\mathcal{Y}^k-\mathcal{G}^{k+1})-\nabla_{\mathcal{G}}H(\mathcal{Y}^k,A_{(N)}^k)\|_F,\\ &\leq \|\nabla_{\mathcal{G}}H(\mathcal{G}^{k+1},A_{(N)}^k)-\nabla_{\mathcal{G}}H(\mathcal{Y}^k,A_{(N)}^k)\|_F\\ &+\|\frac{1}{\sigma_G^k}(\mathcal{Y}^k-\mathcal{G}^{k+1})\|_F\end{aligned}$$

$$\leq (L_{\nabla_{\mathcal{G}^k} H} + \frac{1}{\sigma_G^k})\|\mathcal{G}^{k+1} - \mathcal{Y}^k\|_F. \tag{23}$$

Similarly, we also have

$$\begin{aligned}
\|f_i^{k+1}\|_F &= \|\nabla_{A_i} H(\mathcal{G}^{k+1}, \{A_j^{k+1}\}_{j=1}^{i}, \{A_j^k\}_{j=i+1}^{N}) + \frac{1}{\sigma_i^k}(Y_i^k - A_i^{k+1}) \\
&\quad - \nabla_{A_i} H(\mathcal{G}^{k+1}, \{A_j^{k+1}\}_{j=1}^{i-1}, Y_i^k, \{A_j^k\}_{j=i+1}^{N})\|_F, \\
&\leq \|\nabla_{A_i} H(\mathcal{G}^{k+1}, \{A_j^{k+1}\}_{j=1}^{i}, \{A_j^k\}_{j=i+1}^{N}) - \nabla_{A_i} H(\mathcal{G}^{k+1}, \{A_j^{k+1}\}_{j=1}^{i-1}, Y_i^k, \{A_j^k\}_{j=i+1}^{N})\|_F \\
&\quad + \|\frac{1}{\sigma_i^k}(A_i^{k+1} - Y_i^k)\|_F \\
&\leq (L_{\nabla_{A_i^k} H} + \frac{1}{\sigma_i^k})\|A_i^{k+1} - Y_i^k\|_F.
\end{aligned} \tag{24}$$

Thus we infer

$$\|f_G^{k+1}\|_F + \|\{f_i^{k+1}\}_{i=1}^{N}\|_F \leq \rho_b \|z^{k+1} - c^k\|_F, \tag{25}$$

where $\rho_b = \max(\frac{1}{\sigma_G^k} + L_{\nabla_{\mathcal{G}^k} H}, \{\frac{1}{\sigma_i^k} + L_{\nabla_{A_i^k} H}\}_{i=1}^{N})$. □

8.4 proof of Theorem 3

Proof (i) Observe that z' can be viewed as an intersection of compact sets

$$z' = \cap_{q \in \mathbb{N}} \overline{\cup_{k \geq q} \{z^k\}}.$$

From proposition 1, z' is compact.

(ii) $\forall \overline{z} \in z'$, there exists a subsequence z^{k_j} such that

$$\lim_{j \to \infty} z^{k_j} = \overline{z}.$$

Let

$$F(z^{k_j}) = \sum_{i=1}^{N} F_i(A_i^{k_j}) + F_G(\mathcal{G}^{k_j}). \tag{26}$$

Since F_i and F_G are lower semicontinuous [23], from Definition 3 and Eq. (26), we obtain that

$$\lim_{j \to \infty} \inf F(z^{k_j}) \geq F(\overline{z}). \tag{27}$$

Choosing $k = k_j - 1$, from Eq. (7), we infer

$$\begin{aligned}
\lim_{j \to \infty} \sup F_G(\mathcal{G}^{k_j}) &\leq \lim_{j \to \infty} \sup(F_G(\overline{\mathcal{G}}) + \frac{1}{2\sigma_G^k}\|\overline{\mathcal{G}} - \mathcal{Y}^k\|_F^2 \\
&\quad + \langle \nabla_{\mathcal{G}} H(\mathcal{Y}^k, A_{(N)}^k), \overline{\mathcal{G}} - \mathcal{Y}^k \rangle), \\
\lim_{j \to \infty} \sup F_i(A_i^{k_j}) &\leq \lim_{j \to \infty} \sup(F_i(\overline{A_i}) + \frac{1}{2\sigma_i^k}\|\overline{A_i} - Y_i^k\|_F^2 \\
&\quad + \langle \nabla_{A_i} h_i(Y_i^k), \overline{A_i} - Y_i^k \rangle),
\end{aligned} \tag{28}$$

where $h_i(Y_i^k) = H(\mathcal{G}^{k+1}, \{A_n^{k+1}\}_{n=1}^{i-1}, Y_i^k, \{A_n^k\}_{n=i+1}^{N})$, $\lim_{j \to \infty} \mathcal{G}^{k_j} = \overline{\mathcal{G}}$, $\lim_{j \to \infty} A_i^{k_j} = \overline{A_i}$. Since $\lim_{j \to \infty} z^{k_j} = \overline{z}$ and $\lim_{j \to \infty} \|z^{k_j} - c^{k_j - 1}\|_F = 0$ (Theorem 2(ii)), we can get that

$$\lim_{j \to \infty} \|\overline{A_i} - Y_i^k\|_F \leq \lim_{j \to \infty} \|\overline{A_i} - A_i^{k_j}\|_F + \lim_{j \to \infty} \|A_i^{k_j} - Y_i^{k_j - 1}\|_F$$

$$= 0,$$

$$\lim_{j\to\infty} \|\overline{\mathcal{G}} - \mathcal{Y}^k\|_F \leq \lim_{j\to\infty} \|\overline{\mathcal{G}} - \mathcal{G}^{k_j}\|_F + \lim_{j\to\infty} \|\mathcal{G}^{k_j} - \mathcal{Y}^{k_j-1}\|_F$$
$$= 0. \tag{29}$$

From Eq. (28) and Eq. (29), we infer

$$\lim_{j\to\infty} \sup F_i(A_i^{k_j}) \leq \lim_{j\to\infty} \sup F_i(\overline{A_i}),$$
$$\lim_{j\to\infty} \sup F_G(\mathcal{G}^{k_j}) \leq \lim_{j\to\infty} \sup F_G(\overline{\mathcal{G}}), \tag{30}$$

where $\lim_{j\to\infty} \mathcal{G}^{k_j} = \overline{\mathcal{G}}$, $\lim_{j\to\infty} A_i^{k_j} = \overline{A_i}$. Therefore, from Eq. (26) and Eq. (30), we have

$$\lim_{j\to\infty} \sup F(z^{k_j}) \leq \lim_{j\to\infty} \sup F(\overline{z}). \tag{31}$$

From Eq. (26), Eq. (31) and Eq. (27), we infer

$$\lim_{j\to\infty} F(z^{k_j}) = F(\overline{z}). \tag{32}$$

Since Theorem 2, we have

$$\lim_{j\to\infty} H(z^{k_j}) = H(\overline{z}). \tag{33}$$

Thus from Eq. (32) and Eq. (33), we infer

$$\lim_{j\to\infty} H(z^{k_j}) + \lim_{j\to\infty} F(z^{k_j}) = \lim_{j\to\infty} (H(z^{k_j}) + F(z^{k_j}))$$
$$= \lim_{j\to\infty} J(z^{k_j})$$
$$= J(\overline{z}).$$

This means J is constant on z'.

(iii) From Eq. (20) and Eq. (22), we know that

$$f_G^{k+1} \in \partial_{\mathcal{G}} J(\mathcal{G}^{k+1}, A_{(N)}^k),$$
$$f_i^{k+1} \in \partial_{A_i} J(\mathcal{G}^{k+1}, \{A_j^{k+1}\}_{j=1}^i, \{A_j^k\}_{j=i+1}^N), \tag{34}$$

where the definitions of f_i^{k+1} and f_G^{k+1} are the same as in Eq. (8). From Theorem 2(ii) and Lemma 1, we infer

$$\lim_{k\to\infty} (\|\{f_i^{k+1}\}_{i=1}^N\|_F + \|f_G^{k+1}\|_F) \leq \lim_{k\to\infty} \rho_b \|z^{k+1} - c^k\|_F$$
$$= 0. \tag{35}$$

Let

$$z_1^k = \mathcal{G}^k, z_{i+1}^k = A_i^k. \tag{36}$$

From Theorem 2(i), we know that

$$\lim_{k\to\infty} J(\left\{z_i^{k+1}\right\}_{i=1}^{j-1}, z_j^{k+1}, \left\{z_i^k\right\}_{i=j+1}^{N+1}) = \lim_{k\to\infty} J(z^k)$$
$$= J(z^*). \tag{37}$$

Therefore, from Eq. (34), Eq. (35), Eq. (36) and Eq. (37), we have

$$0 \in \partial_{z_j} \lim_{k\to\infty} J(\left\{z_i^{k+1}\right\}_{i=1}^{j-1}, z_j^{k+1}, \left\{z_i^k\right\}_{i=j+1}^{N+1})$$
$$= \partial_{z_j} J(z^*).$$

This means $0 \in \partial J(z^*)$. □

8.5 proof of Theorem 4

Proof From Theorem 1, we know that $\sum_{q=1}^{Q} \frac{\alpha_q}{2} \operatorname{Tr}((A_N)^T L_q A_N)$ is an infinitely differentiable function, and the norm of its derivatives of any order is also continuous, so $\sum_{q=1}^{Q} \frac{\alpha_q}{2} \operatorname{Tr}((A_N)^T L_q A_N)$ is a real analytic function (real analytic functions are all KŁ functions [33]). Since the Frobenius norm, ℓ_p-norm is also all KŁ functions [23], it follows that Eq. (4) is a KŁ function. Therefore, from Definition 6, there exists a concave function ϕ so that

$$\phi^{'}(J(z^k) - J(\overline{z}))dist(0, \partial J(z^k)) \geq 1. \tag{38}$$

From ϕ is the convex function, we have

$$\begin{aligned}\phi(J(z^{k+1}) - J(\overline{z})) \leq \phi(J(z^k) - J(\overline{z})) \\ + \phi^{'}(J(z^k) - J(\overline{z}))(J(z^{k+1}) - J(z^k)).\end{aligned} \tag{39}$$

From Lemma 1 and Theorem 3, we infer

$$dist(0, \partial J(z^k)) \leq \rho_b \|z^k - c^{k-1}\|_F. \tag{40}$$

Since the Eq. (38) and Eq. (40), we infer

$$\begin{aligned}\phi^{'}(J(z^k) - J(\overline{z})) &\geq \frac{1}{dist(0, \partial J(z^k))} \\ &\geq \frac{1}{\rho_b \|z^k - c^{k-1}\|_F}.\end{aligned} \tag{41}$$

Let $J(k) = J(z^k) - J(\overline{z})$, from Eq. (39), Eq. (40), Eq. (41) and Theorem 2(i), we have

$$\begin{aligned}\phi(J(k)) - \phi(J(k+1)) &\geq \phi^{'}(J(k))(J(k) - J(k+1)) \\ &\geq \frac{J(k) - J(k+1)}{\rho_b \|z^k - c^{k-1}\|_F} \\ &\geq \frac{\rho \|z^{k+1} - c^k\|_F^2}{\rho_b \|z^k - c^{k-1}\|_F}.\end{aligned}$$

Define C=$\frac{\rho}{\rho_b}$, C is a constant, so we infer

$$\|z^{k+1} - c^k\|_F^2 \leq C(\phi(J(k)) - \phi(J(k+1)))\|z^k - c^{k-1}\|_F.$$

Using the fact that $2ab \leq a^2 + b^2$

$$2\|z^{k+1} - c^k\|_F \leq C(\phi(J(k)) - \phi(J(k+1))) + \|z^k - c^{k-1}\|_F.$$

Sum both sides

$$\begin{aligned}2\sum_{k=l+1}^{K} \|z^{k+1} - c^k\|_F \leq \sum_{k=l+1}^{K} \|z^k - c^{k-1}\|_F \\ + C(\phi(J(l+1)) - \phi(J(K+1))) \\ = C(\phi(J(l+1)) - \phi(J(K+1))) \\ + \|z^{l+1} - c^l\|_F + \sum_{k=l+1}^{K} \|z^{k+1} - c^k\|_F.\end{aligned} \tag{42}$$

From Eq. (42), we can get that

$$\lim_{K\to\infty} \sum_{k=l+1}^{K} \|z^{k+1} - c^k\|_F \leq \|z^{l+1} - c^l\|_F + C\phi(J(l+1))$$

$$
\begin{aligned}
&- \lim_{K\to\infty} C\phi(J(K+1)) \\
&< \infty.
\end{aligned}
$$

Thus we have

$$
\sum_{k=l+1}^{\infty} \|z^{k+1} - c^k\|_F < \infty. \tag{43}
$$

From Algorithm 1, no matter $c^k = z^k + \beta_k(z^k - z^{k-1})$ or $c^k = z^k$, we always have

$$
\|z^{k+1} - z^k\|_F - \beta_k\|z^k - z^{k-1}\|_F \ \leq \|z^{k+1} - c^k\|_F. \tag{44}
$$

From Eq. (43) and Eq. (44), we know that

$$
\begin{aligned}
\sum_{k=l+1}^{\infty} (\|z^{k+1} - z^k\|_F - \beta_k\|z^k - z^{k-1}\|_F) &\leq \sum_{k=l+1}^{\infty} \|z^{k+1} - c^k\|_F \\
&< \infty.
\end{aligned}
$$

Since

$$
\|z^{k+1} - z^k\|_F - \beta_{max}\|z^k - z^{k-1}\|_F \leq \|z^{k+1} - z^k\|_F - \beta_k\|z^k - z^{k-1}\|_F,
$$

we also have

$$
\begin{aligned}
\sum_{k=l+1}^{\infty} (\|z^{k+1} - z^k\|_F - \beta_{max}\|z^k - z^{k-1}\|_F) &\leq \sum_{k=l+1}^{\infty} (\|z^{k+1} - z^k\|_F - \beta_k\|z^k - z^{k-1}\|_F) \\
&< \infty,
\end{aligned} \tag{45}
$$

and

$$
\begin{aligned}
&\sum_{k=l+1}^{\infty} \|z^{k+1} - z^k\|_F - \sum_{k=l+1}^{\infty} \beta_{max}\|z^k - z^{k-1}\|_F \\
&= \sum_{k=l+1}^{\infty} (\|z^{k+1} - z^k\|_F - \beta_{max}\|z^{k+1} - z^k\|_F) + \beta_{max}\|z^{l+1} - z^l\|_F \\
&= \sum_{k=l+1}^{\infty} (1 - \beta_{max})\|z^{k+1} - z^k\|_F + \beta_{max}\|z^{l+1} - z^l\|_F.
\end{aligned} \tag{46}
$$

Let $s^k = \|z^{k+1} - z^k\|_F$, from Eq. (45) and Eq. (46), we have

$$
\begin{aligned}
\sum_{k=l+1}^{\infty} (1 - \beta_{max})s^k - \beta_{max}s^l &= \sum_{k=l}^{\infty}(s^{k+1} - \beta_{max}s^k) \\
&\leq \sum_{k=l+1}^{\infty} \|z^{k+1} - c^k\|_F \\
&< \infty.
\end{aligned}
$$

Thus we infer

$$
\sum_{k=l+1}^{\infty} (1 - \beta_{max})s^k < \infty. \tag{47}
$$

From $(1-\beta_{max})$ is a positive constant and Eq. (47), we infer

$$\sum_{k=l+1}^{\infty} \|z^{k+1} - z^k\|_F = \sum_{k=l+1}^{\infty} s^k < \infty.$$

Thus we have

$$\sum_{k=0}^{\infty} \|z^{k+1} - z^k\|_F = \sum_{k=0}^{\infty} s^k < \infty.$$

This shows that

$$\begin{aligned}
\lim_{K\to\infty} \|z^{K+p} - z^K\|_F &= \lim_{K\to\infty} \| \sum_{k=K+1}^{K+p} (z^{k+1} - z^k)\|_F \\
&\leq \lim_{K\to\infty} \sum_{k=K+1}^{K+p} \|z^{k+1} - z^k\|_F \\
&= \lim_{K\to\infty} \sum_{k=K+1}^{\infty} s^k \\
&= 0. \qquad (48)
\end{aligned}$$

This means the Theorem 4 is true. □

8.6 proof of Theorem 5

Proof From Appendix B of [11] and Section 3.5 of [12], we know that the computational cost of $\nabla_{A_i} H(\mathcal{G}, A_{(N)})$ is

$$\mathcal{O}(\sum_{j=1}^{N}(\sum_{i=1}^{j} I_i)(\sum_{i=j}^{N} r_i)). \qquad (49)$$

Similarly, the computational cost of $\nabla_{\mathcal{G}} H(\mathcal{G}, A_{(N)})$ is

$$\mathcal{O}(\sum_{j=1}^{N}(\sum_{i=1}^{j} r_i)(\sum_{i=j}^{N} I_i)). \qquad (50)$$

The cost of computing the Lipschitz constant, projection to nonnegative, and tensor unfolding is negligible compared to the cost of computing partial gradients $\nabla_{\mathcal{G}} H(\mathcal{G}, A_{(N)})$ and $\nabla_{A_j} H(\mathcal{G}, A_{(N)})$. Therefore, from Eq. (49) and Eq. (50), the time complexity of the Algorithm 1 in each iteration is approximately estimated as follows.

$$\mathcal{O}(\sum_{j=1}^{N}(\sum_{i=1}^{j} r_i)(\sum_{i=j}^{N} I_i) + N(\sum_{j=1}^{N}(\sum_{i=1}^{j} I_i)(\sum_{i=j}^{N} r_i))).$$

□